

DSC 465
HOMEWORK 1

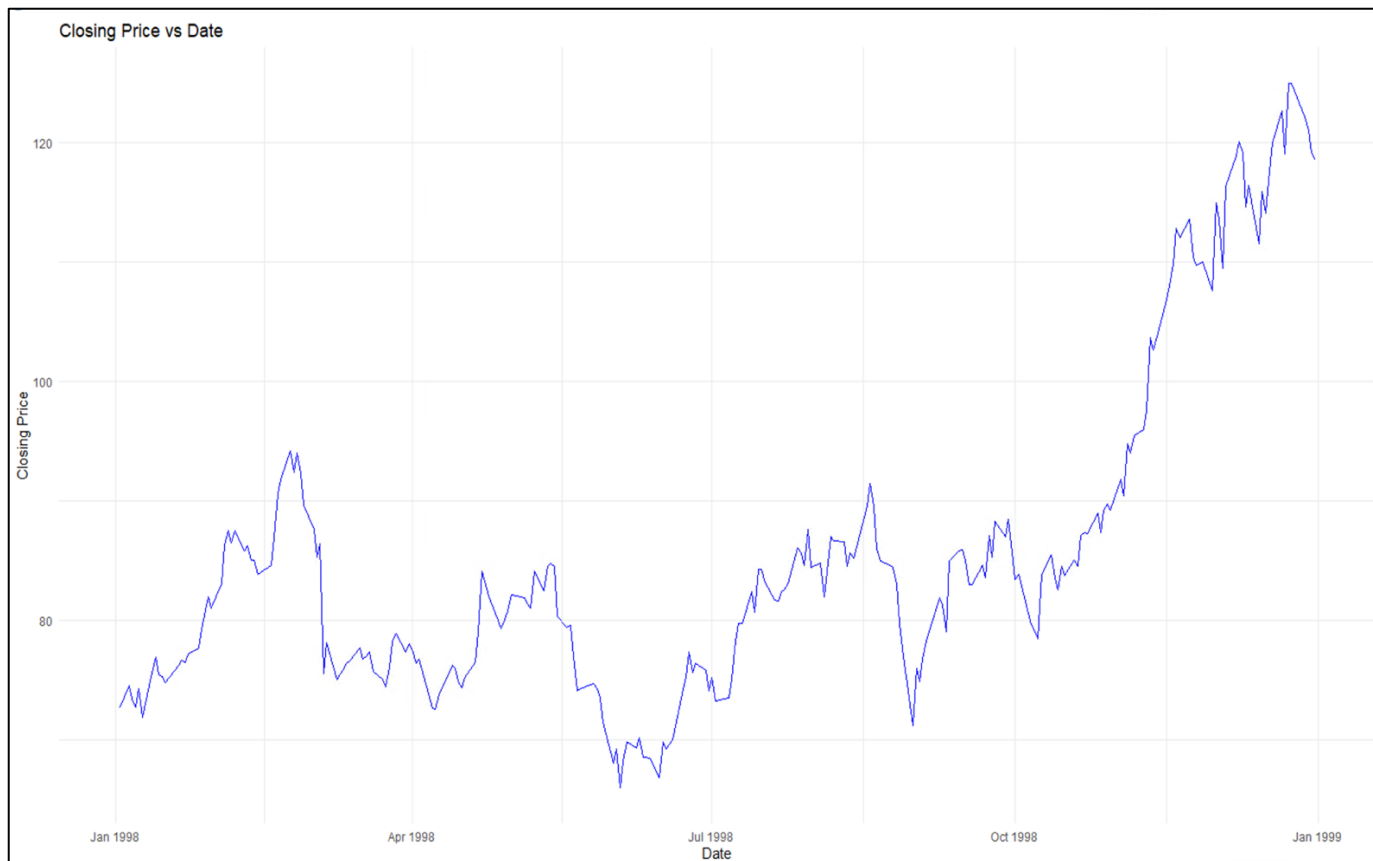
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1) (20 pts) For this problem, we'll look at data about Intel stock (Intel-1998 dataset from the website). The data covers stock market trading for the Intel corporation in 1998. Each row is a day, with the following columns: Date, Trading Day (integer day number, including skips), Open (price at market open), High (highest price of day), Low (lowest price of day), Close (price at market close), Volume (shares traded), and Adj. Close (adjusted closing price, meaning accounting for stock splits, which are not a problem in this data).

Make the specified graphs in either R or Tableau:

- a. Graph the closing price vs. the date with an ordinary line graph. If you use Tableau, you need to right-click on the Date and choose Exact Date from the dropdown menu so that it uses the full date with "day".

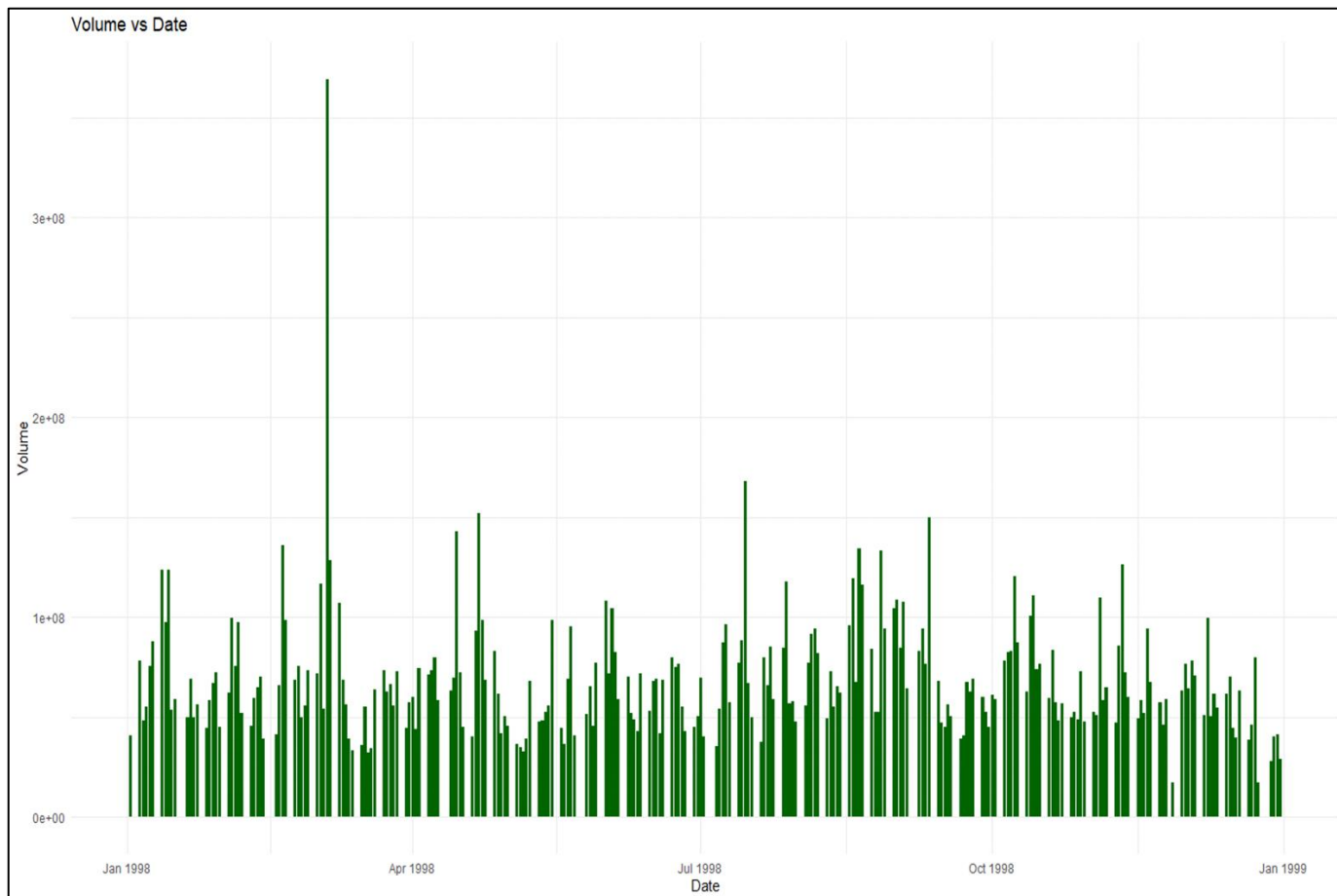
```
> rm(list=ls())
> intel_data <- read.csv(file.choose(),header=TRUE)
> library(ggplot2)
> intel_data$Date <- as.Date(intel_data$Date, format = "%m/%d/%Y")
> str(intel_data)
'data.frame': 252 obs. of 8 variables:
 $ Date      : Date, format: "1998-01-02" "1998-01-05" ...
 $ Trading.Day: int  1 4 5 6 7 8 11 12 13 14 ...
 $ Open      : num  70.7 73.1 73.9 72.8 72.2 ...
 $ High      : num  72.6 75.1 74.3 73.6 74.8 ...
 $ Low       : num  70.5 72 72.7 71.6 72.1 ...
 $ Close     : num  72.6 74.5 73.1 72.8 74.3 ...
 $ Volume    : int  40927200 78368400 48313600 55380800 75730400 88028800 123747600 97348000 123962000
53458400 ...
 $ Adj.Close  : num  16.9 17.4 17.1 17 17.3 ...
> head(intel_data)
      Date Trading.Day Open High Low Close  volume Adj.Close
1 1998-01-02           1 70.69 72.62 70.50 72.62 40927200    16.94
2 1998-01-05           4 73.06 75.14 72.00 74.50 78368400    17.37
3 1998-01-06           5 73.87 74.31 72.69 73.12 48313600    17.05
4 1998-01-07           6 72.75 73.62 71.56 72.75 55380800    16.97
5 1998-01-08           7 72.25 74.81 72.12 74.31 75730400    17.33
6 1998-01-09           8 74.56 75.00 71.23 71.87 88028800    16.76
> # Line graph: Closing Price vs Date
> ggplot(intel_data, aes(x = Date, y = Close)) +
+   geom_line(color = "blue") +
+   labs(title = "Closing Price vs Date", x = "Date", y = "Closing Price") +
+   theme_minimal()
> |
```



Interpretation: The stock's closing price showed a clear upward trend throughout 1998. Price fluctuations are visible, especially between March and October, where the stock underwent several rises and dips. The upward trend towards the end of the year could indicate positive market sentiment or external economic factors influencing Intel's stock.

b. Graph the Volume vs. the exact Date as in the last part with a bar graph.

```
> #Bar Graogh: Volume vs Date
> ggplot(intel_data, aes(x = Date, y = Volume)) +
+   geom_bar(stat = "identity", fill = "darkgreen") +
+   labs(title = "Volume vs Date", x = "Date", y = "Volume") +
+   theme_minimal()
```



Interpretation: This bar chart shows a time series analysis with Volume on the y-axis and Date on the x-axis. The chart spans from January 1998 to January 1999 with a few prominent spikes in volume indicating certain days had significantly higher volumes than others. The trend mostly shows consistent volumes over time, except for the few major spikes which could represent events like promotions or high-sales days.

c. Create a scatterplot that graphs the Volume on the x-axis and the daily price range on the y-axis. You will need to create an additional column that contains the "range" of the prices for the day as the difference between the fields High and Low.

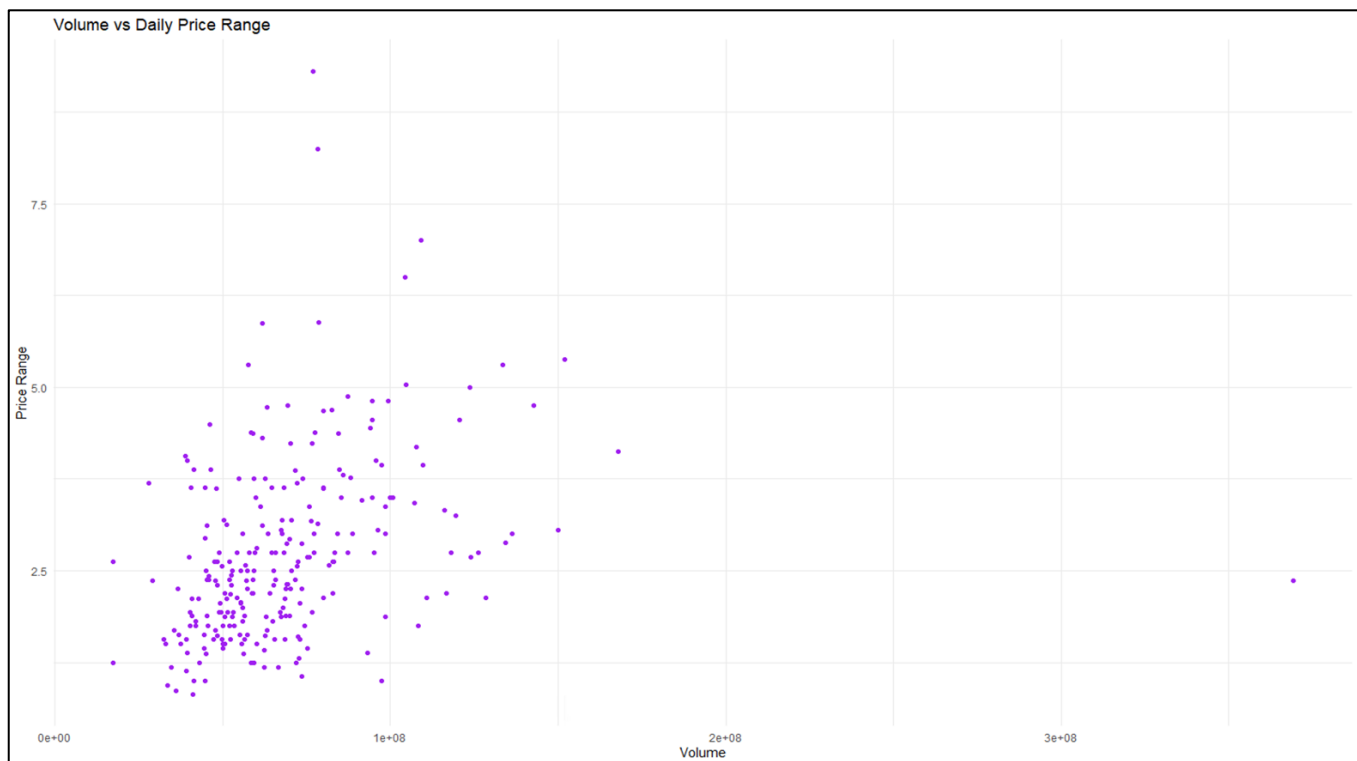
$$\text{Range} = \text{High} - \text{Low}$$

Tableau can do it with a Calculated Field. In R you can do it by making a new column equal to the result from subtracting the two columns. In Tableau, to get a scatter plot, you will need to right click on both the Range and Volume entries in graph and change them to "Dimensions".

```

> intel_data$Range <- intel_data$High - intel_data$Low
> view(intel_data)
> # Scatterplot: Volume vs Price Range
> ggplot(intel_data, aes(x = Volume, y = Range)) +
+   geom_point(color = "purple") +
+   labs(title = "Volume vs Daily Price Range", x = "Volume", y = "Price Range") +
+   theme_minimal()

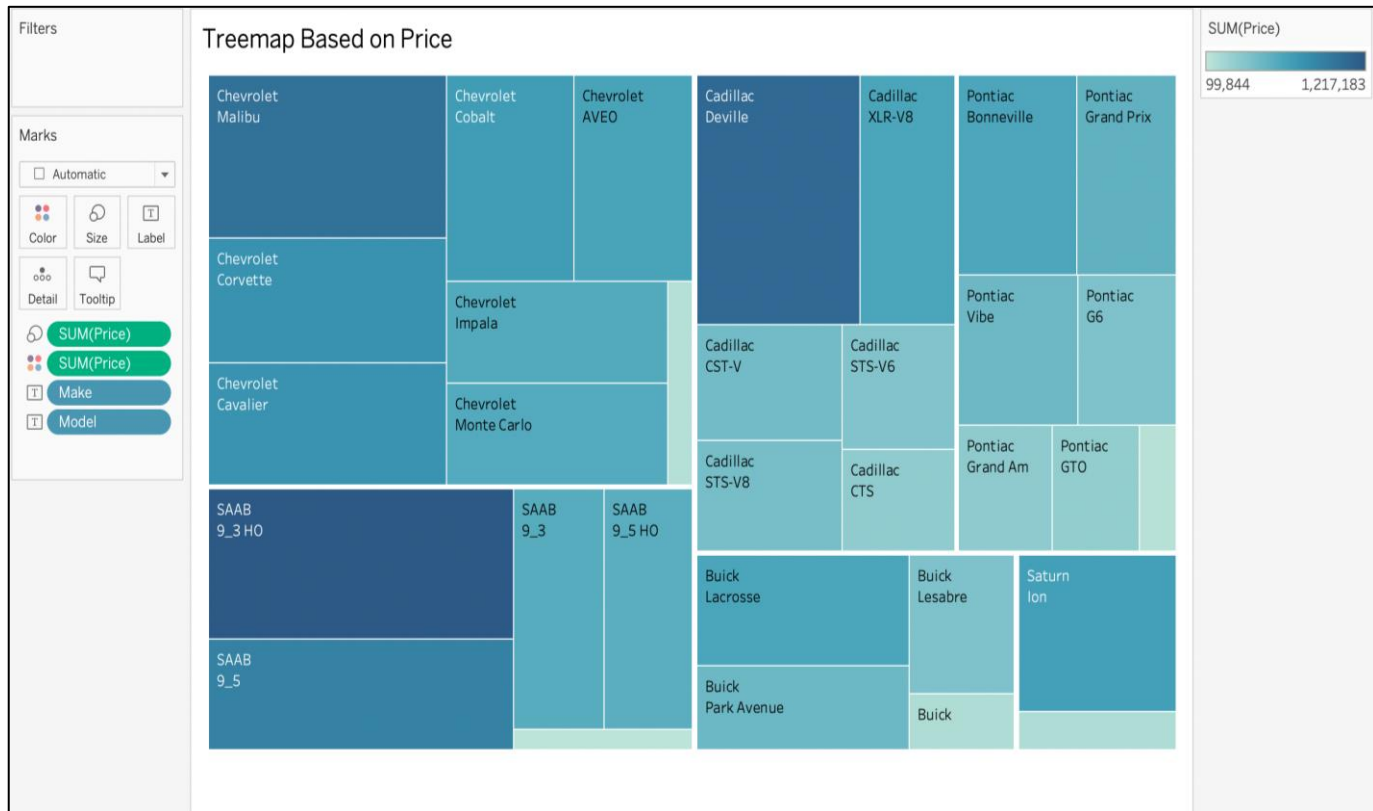
```



Interpretation: The scatter plot reveals that most trading days had moderate volumes, clustered between 0 and 1e+08 shares traded and price ranges between 2.5 and 5.0. There are some outliers indicating unusually high-volume days with moderate price ranges. There seems to be no strong correlation between the number of shares traded and the range of daily prices.

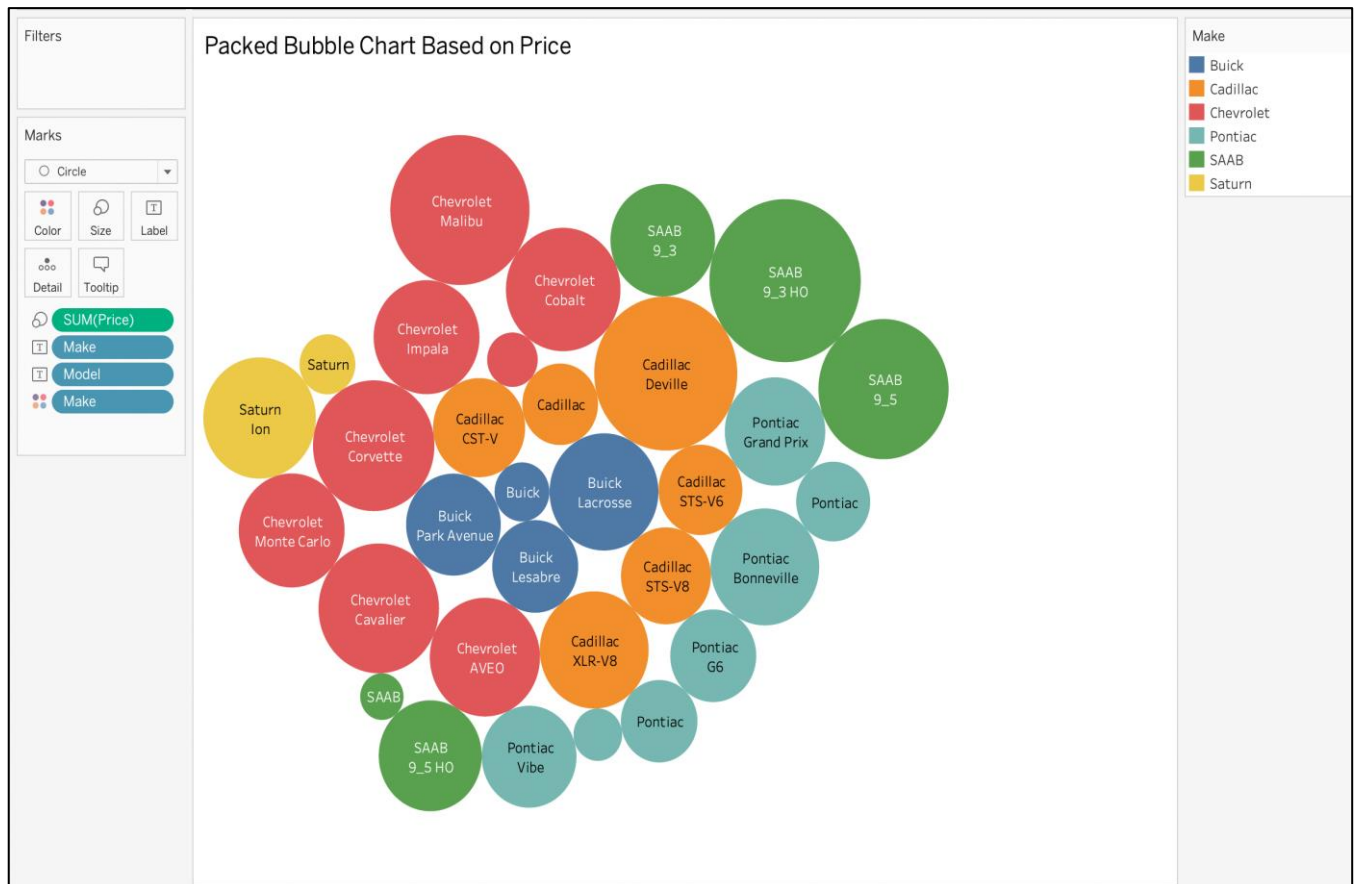
2) (20 pts) Use Tableau for this question. Open the GM cars dataset included with this assignment (gmcar_price.txt). Each row represents a different car that was sold and includes information about features like the mileage and the price of sale. Hint: use the “Show Me” menu.

a. A tree map based on Price with a main subdivision for the Make of the car and a minor subdivision based on the Model. Because each row of the data file represents a single car but each box in the tree map represents all the cars with a given make and model, pay very close attention to what kind of aggregation is being used.



Interpretation: This above tree map visualizes different car models and their prices. Each block represents a different car model, sized according to the total price (sum of prices). Cadillac Deville, Chevrolet Malibu and Pontiac are larger which means they represent higher total prices or a larger number of cars sold at higher prices. The color gradient from light to dark blue indicates price ranges where darker shades represent higher prices.

b. A packed bubble chart of the same type.



Interpretation: The above chart shows a Packed Bubble Chart where the size of each bubble corresponds to the price associated with different car brands. Brands like SAAB and Chevrolet have larger bubbles indicating higher prices while smaller bubbles like Buick and Saturn represent lower-priced cars. The bubbles are color-coded by Make with SAAB in green, Cadillac in orange and Chevrolet in red.

c. Write a short paragraph discussing the differences between the two plots. Describe for each something that displayed more clearly than with the other.

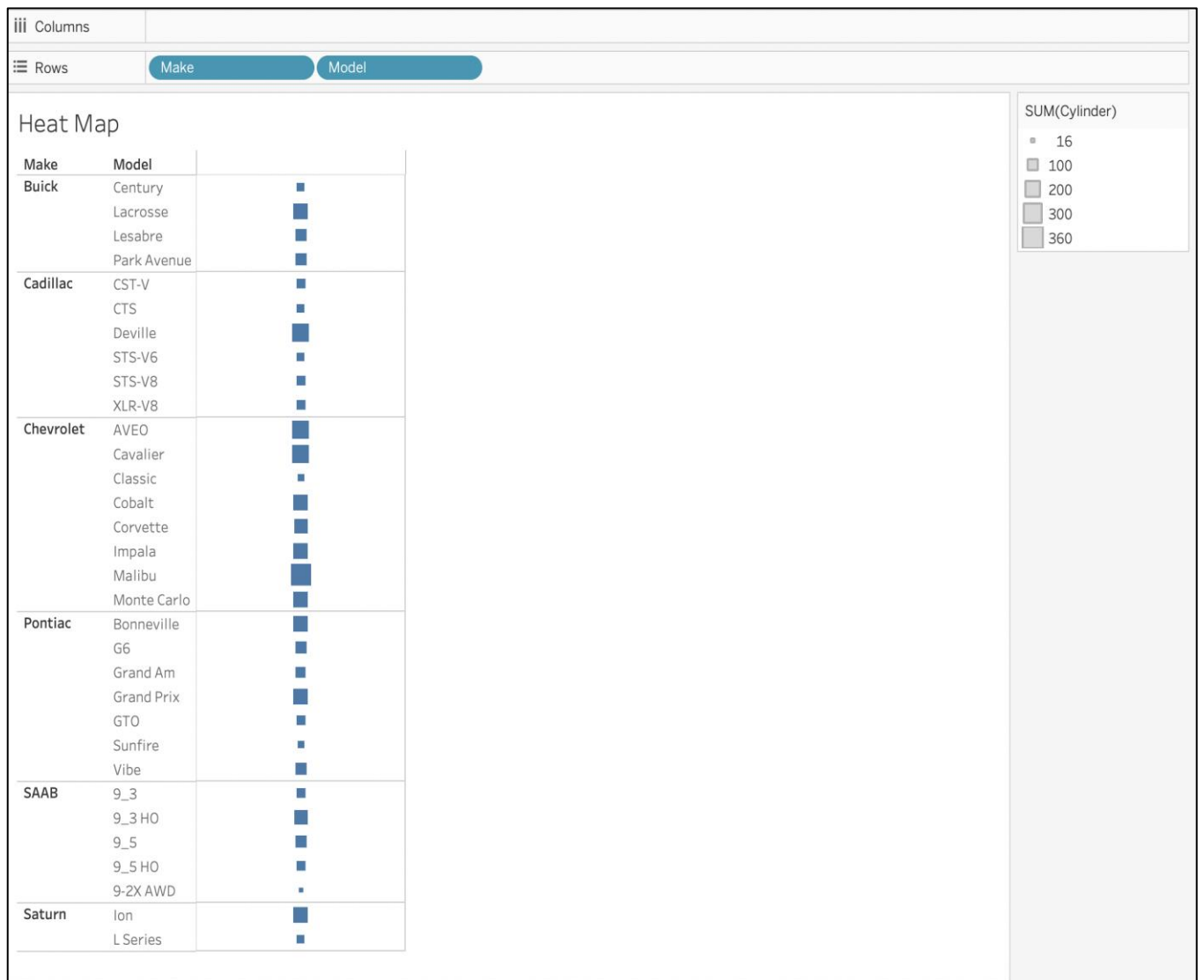
Ans: Tree map:

- Better suited for comparing relative prices between different makes and models.
- Offers a clearer subdivision between makes and models and helps users quickly understand how each model contributes to the overall price.

Packed Bubble Chart:

- More visually engaging but less informative for precise price comparisons.
- Best used when the focus is on general market dominance of certain makes/models, but not for detailed analysis.

d. Create a contingency plot (Tableau calls it a heat map under Show Me) showing with color the number of cars (Number of Records) of each Type sold by each Make. Explain at least one observation about that data that this chart makes it easy to see.

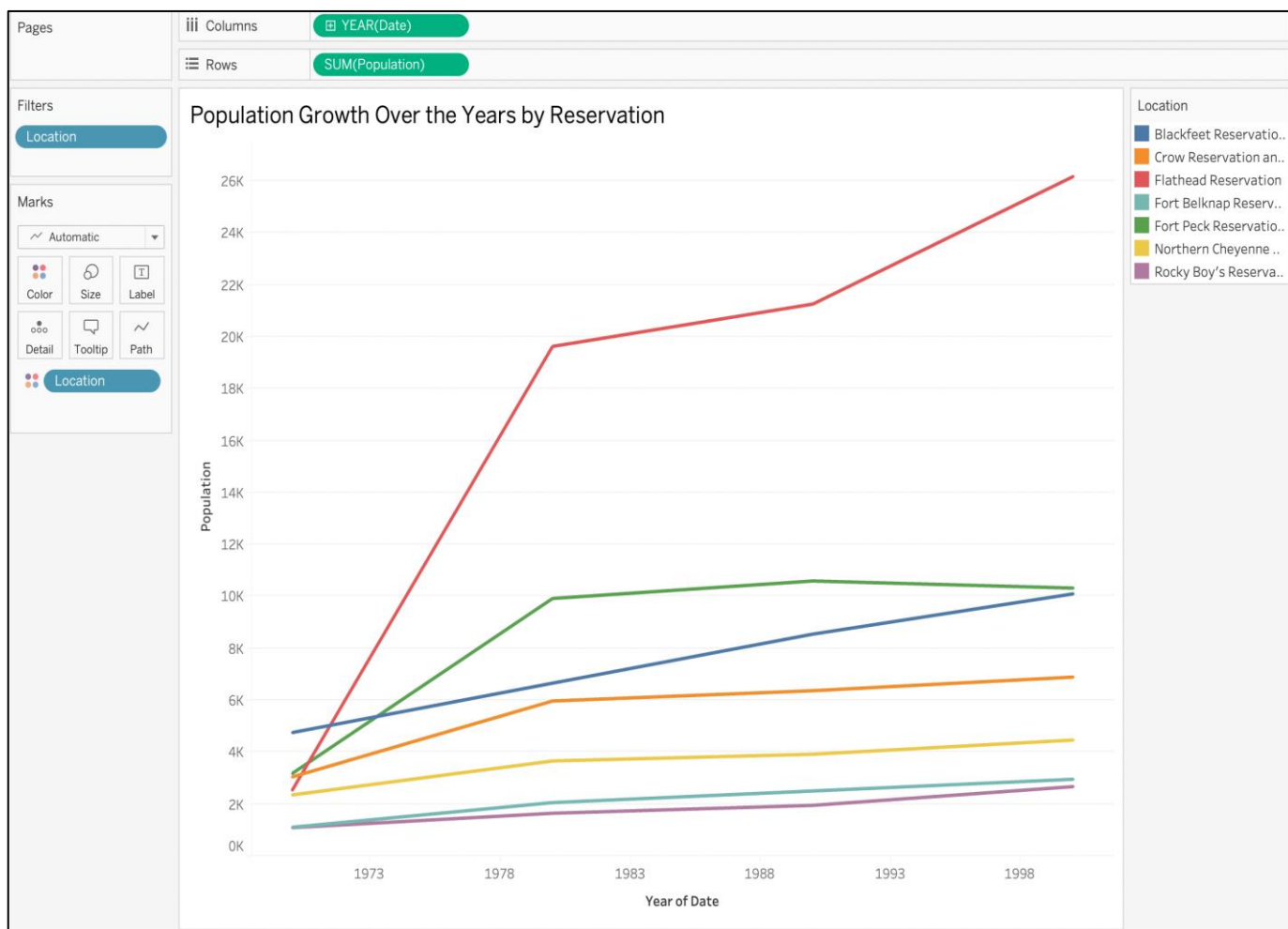


This heat map highlights the number of cars sold by each make and type. Darker colors indicate higher sales allowing quick identification of which car types are most popular for specific manufacturers.

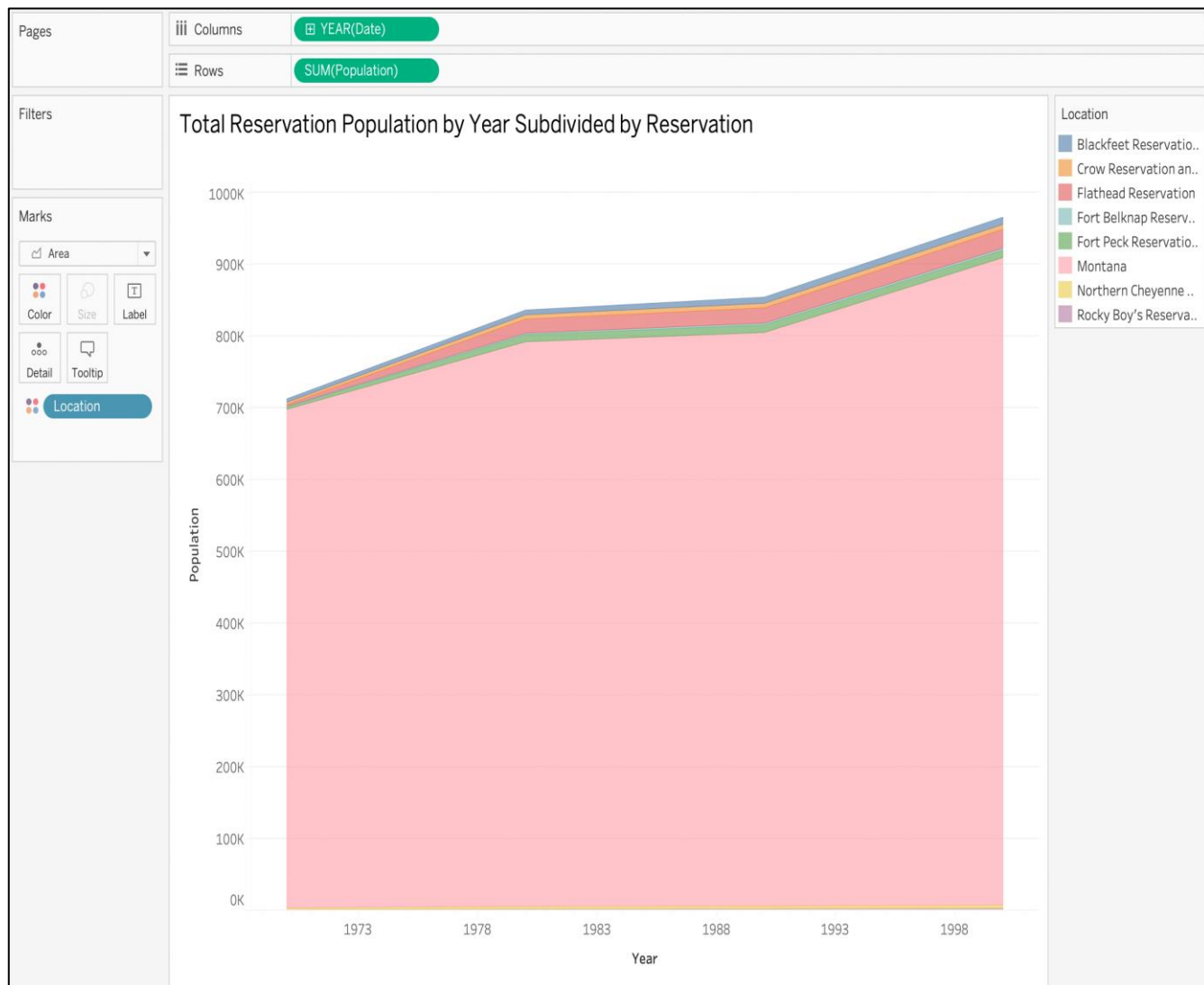
The heat map shows that certain types are predominantly sold by specific manufacturers which may reflect market specialization or consumer preferences.

3. Create graphs to show the following information, using appropriate graph types. Make sure that the graphs are properly labeled and that the axis scales properly reflect the type of data represented.

- a. One chart that graphs the population growth over the years for the individual reservations.



- b. One that graphs the total reservation population for each year, subdivided among the different reservations. The difference between this and (a) is that in (b) we are not looking only at each population individually but at the growth of the total population of all of them together, then subdivided by the reservations.



The stacked area chart visualizes the combined population growth across all reservations. It shows how each reservation contributes to the overall population change over time. Some reservations have grown more steadily than others and their contributions to the overall population growth vary across decades.

4) For this question, answer only with text. You may include an illustration if you would like, but you do not need to visualize data for this question.

a. Explain what we mean by ‘pre-attentive’ attributes. Are these as effectively recognized by human perception when they are used in combinations?

Ans: Pre-attentive Attributes- Pre-attentive attributes refer to visual properties that the human brain can process quickly and effortlessly without conscious thought. These attributes are detected within milliseconds of seeing an image allowing us to immediately grasp the meaning of the data being presented. Examples of pre-attentive attributes include:

- Color
- Position
- Shape
- Size
- Orientation
- Texture
- Length

When used effectively, pre-attentive attributes help viewers instantly identify patterns, trends or outliers in visual data. However, using pre-attentive attributes in combination can reduce their effectiveness. For instance, using too many visual cues (e.g., color and shape changes at once) may overwhelm viewers or cause confusion, making it harder to process the data quickly. Therefore, it's generally better to use them sparingly and focus on one or two attributes to highlight important data points.

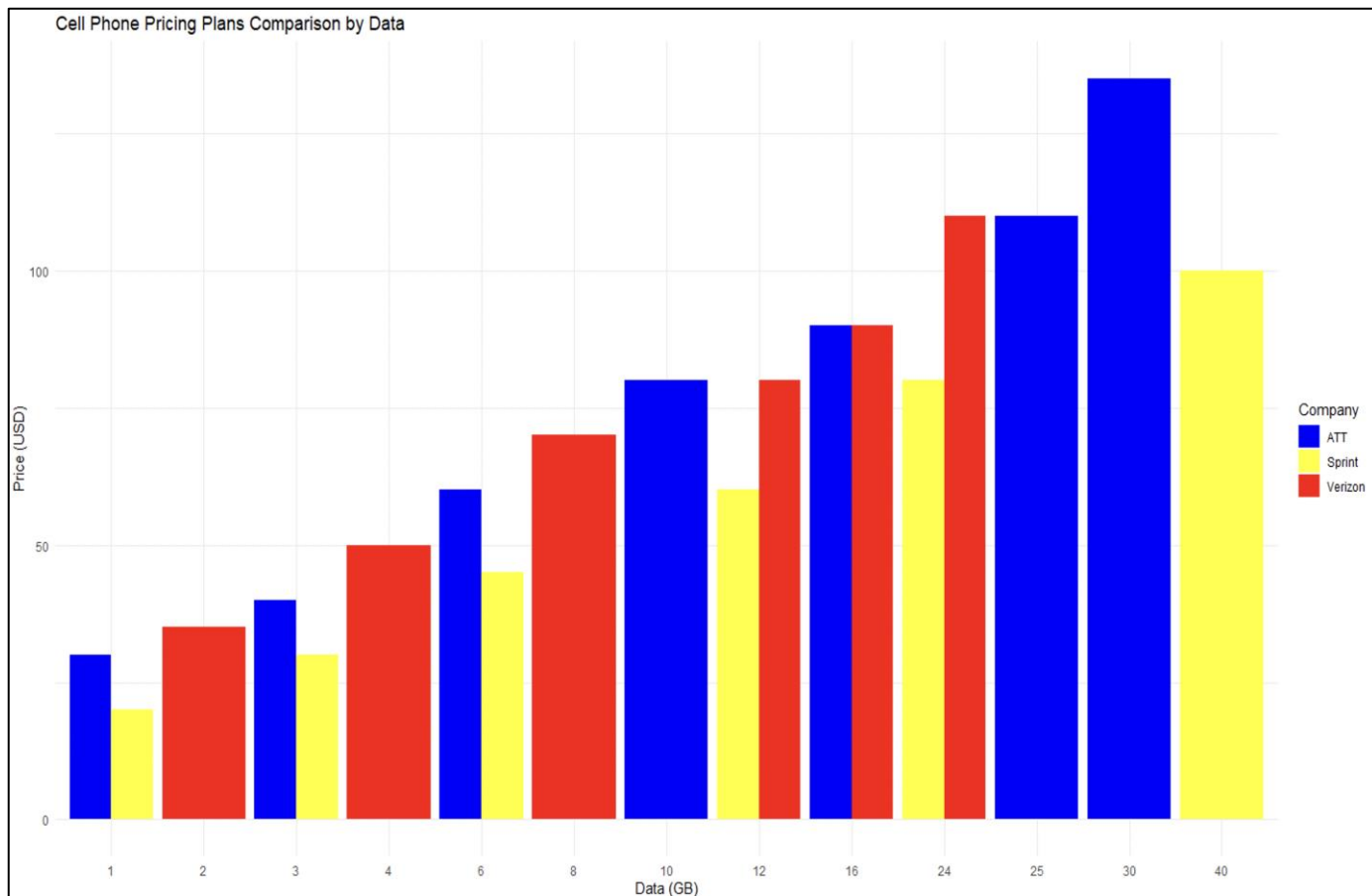
b. Use Weber’s Law to explain why it is important to include 0 in the numerical axis of a bar chart.

Ans: Weber’s Law relates to human perception, stating that the ability to perceive a difference between two stimuli is proportional to the relative difference rather than the absolute difference. This means that humans are better at detecting changes when they are compared to a baseline reference point.

In case of bar charts, including 0 on the numerical axis is important because it provides a clear baseline for comparison. Without 0, the relative size of the bars can be misleading, as viewers might perceive small differences between bars as larger than they actually are. The starting point of 0 anchors the bars and ensures that their length accurately reflects the true proportions of the data preventing visual distortion in comparisons.

5. Comparing Companies Pricing Plans

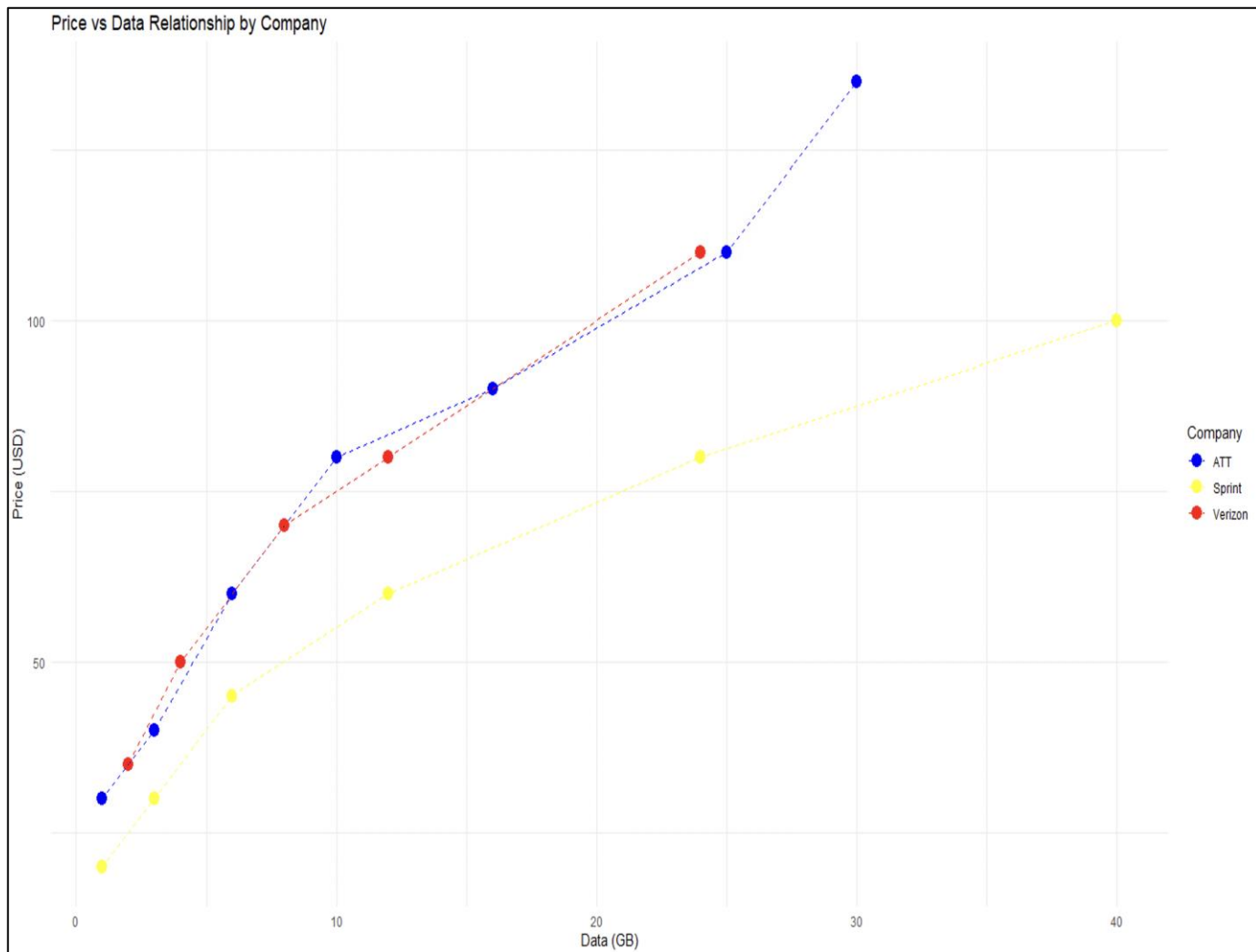
This visualization compares the pricing plans of each company, showing the price difference for different data packages. It aims to help users easily see which company offers the best price for different data amounts.



This bar chart uses **data amounts (GB)** on the x-axis and **price** on the y-axis. The bars are grouped by company, making it easy to compare prices for each data bucket. This visualization is meant to help users see which company offers cheaper or more expensive plans at each data level.

Price vs. Data Relationship Across Companies

This scatter plot visualizes the relationship between the **data offered (GB)** and the **price** for each company, helping users observe trends in pricing as data amounts increase. This plot also helps to compare how pricing strategies differ between companies.



This scatter plot shows the **data amounts (GB)** on the x-axis and the **price** on the y-axis, with each company represented by a different color. Lines connect the points for each company to show trends over increasing data amounts. This visualization makes it easier to identify the general pricing strategy and how the price changes as more data is included in the plan.